rogii-wellbore-geology-prediction

22nd Place Solution — Five Engines and a Hard Lesson About Leaderboards

Rank	22
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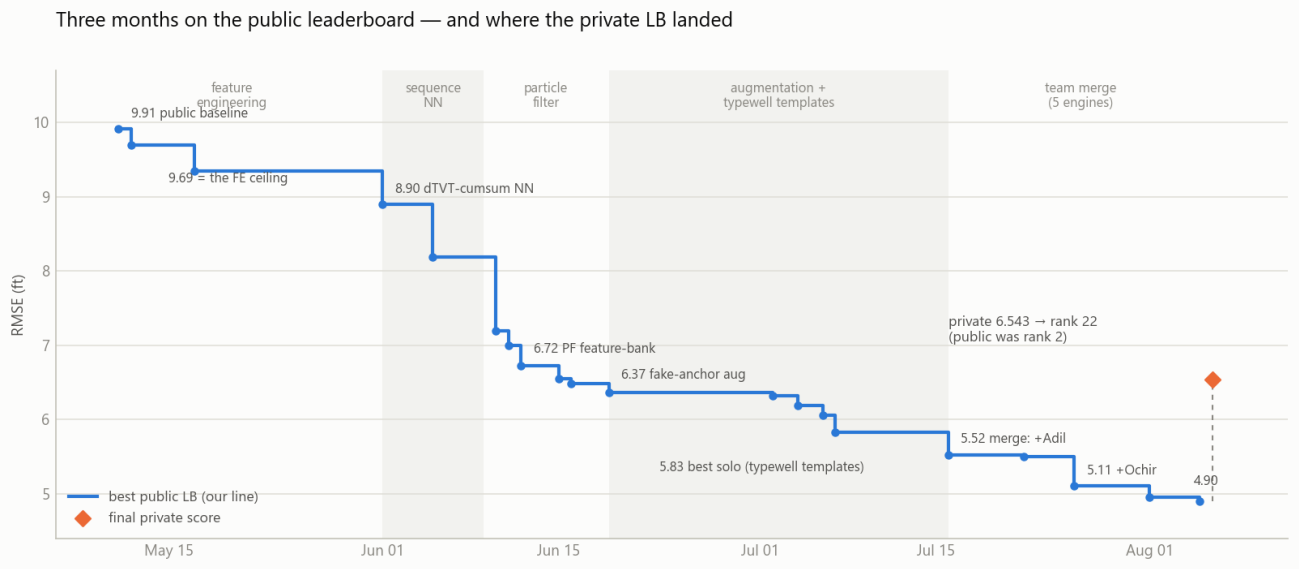
First of all, thanks to ROGII and Kaggle for a genuinely deep problem, and to everyone who shared ideas in the forum.

We finished **public 4.901 (rank 2) → private 6.543 (rank 22)** — the last gold slot, and the biggest public→private drop inside the gold zone. So this writeup is two stories at once: how five separate engines were built and merged, and what the private leaderboard taught us about which of our decisions were real. We'll try to be honest about both.

Team **N & A & O & A & A**: Phong Nguyen Quoc, Adil Khabibullin, Andrey Serzhenko, Ochir Dorzhiev, Alexey Zakharov — a merge of four independent solo lines. The first half of this post is my line; the second half summarizes each teammate's engine.

TL;DR

- **Solo line:** feature engineering capped at LB 9.69 → an anchor-relative **dTVT-cumsum NN** broke the wall (8.90) → **particle filter as a feature factory** (6.72) → **fake-anchor augmentation** (6.37) → **typewell-template forensics + sibling-lateral likelihood tables** (best solo **5.83**).
- **Merge:** 5 engines → a 40–50 member cached-OOF pool, shrink-NNLS + Ridge stack, governed by two hard membership rules (**decode-parity** and **solo-LB-proven**) plus a Q-vs-HMM disagreement gate. Public **4.90**.
- **Private:** all 50 of our submissions scored worse on private than public (**+1.47 ± 0.95 ft**); inside the endgame band the two leaderboards were anti-correlated (Spearman **−0.48**), while our 773-well CV correlated **+0.84** with private. The submission our public-LB evidence told us to reject was our best private score. Trust big-N CV.



Part I — The solo climb: 9.91 → 5.83

#	Change	Public LB
0	Public aeroridge baseline (LGB + CB + TabICL)	9.910

#	Change	Public LB
1	2-D basin priors `p(GR, gr_d1)` per formation — the only FE that survived	9.693
2	**dTVT-cumsum NN** added as a blend member	8.896
3	GPU particle filter → multi-profile PF members	6.997
4	PF reused as **NN input channels** (feature bank)	6.719
5	Ridge residual meta-stack	6.480
6	**Fake-anchor augmentation** member	6.367
7	Template likelihood family + **loose test-side gate**	6.062
8	Symmetric retrain on GR-cluster tables — best solo	**5.826**

The feature-engineering wall

- ~40 versions of FE on top of the public stack; everything except the 2-D basin prior came back `negative` or `overfit_to_cv`.
- Screening lesson: **subset screens lie** — features ranked #1/189 on a 30%-well screen regressed +0.4 at full scale.
- Prior lesson: 1-D amplitude priors `p(GR|F)` overfit the train basin; conditioning on a derivative axis (`gr_d1`) is what made them transfer.
- Real conclusion of the era: **the lever was never features — it was the model class.**

Predict the increment, not the depth (LB 8.90)

TVT is conserved along a lateral — the absolute value is an arbitrary per-well datum; only the relative displacement is learnable:

```
predict per-row dTVT  →  TVT = TVT_anchor + cumsum(dTVT)
```

- Architecture: full-resolution **dilated TCN** (10 blocks, receptive field ≈ 4,093 rows), deliberately **minimal features (14)**.
- Dual loss: global RMSE on the reconstructed cumsum + local SmoothL1 on increments. The local weight was set by **measuring per-term gradient norms** (the cumsum term front-loads gradients ~500×; at the "obvious" weight the local term was numerically inert → λ=30).
- **Finding — buy diversity, not standalone accuracy.** Standalone this NN was barely better than the GBM pool (10.30 vs 10.40), but its residual correlation with the pool was only **0.60** → one shrink-NNLS member took the blend CV down ~1.0 and public LB 9.693 → **8.896**. From here on, every candidate was judged by residual-space correlation and marginal blend lift, never by standalone score.

The particle filter as a feature factory (LB 6.72)

- GPU-ported the PF; multi-profile PF trajectories as members → 6.997.

- **Finding — PF-as-input beats PF-as-member.** Feeding the PF's per-row profile estimates and per-row spread as NN input channels produced a far stronger member (**6.719**): the NN learns when to trust the filter (fixed the PF's in-zone weakness, −0.9 on low-drift rows) instead of us hand-designing a gate.

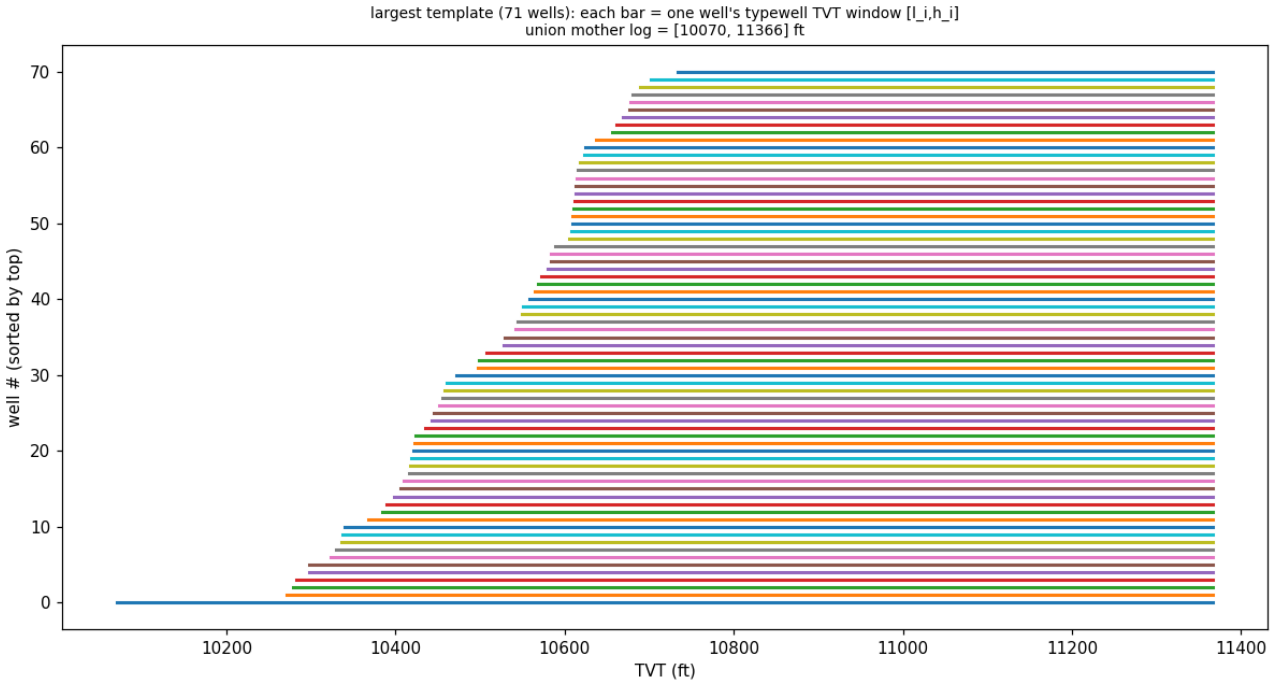
Fake-anchor augmentation (LB 6.37)

- Error audit: ~75% of squared error lived in the far-from-anchor toe — the model simply saw too few far-anchor training examples.
- **Finding — fake-anchor augmentation:** replay each training well with the anchor artificially pulled back (~3× effective labels in the far-anchor regime). One 1-seed member: CV 6.88 → **LB 6.367**, our cleanest CV→LB transfer to that point.
- In hindsight this was the little brother of the synthetic-data factories in the #2/#8/#13 writeups — augmentation was our most under-exploited axis (see What didn't work).

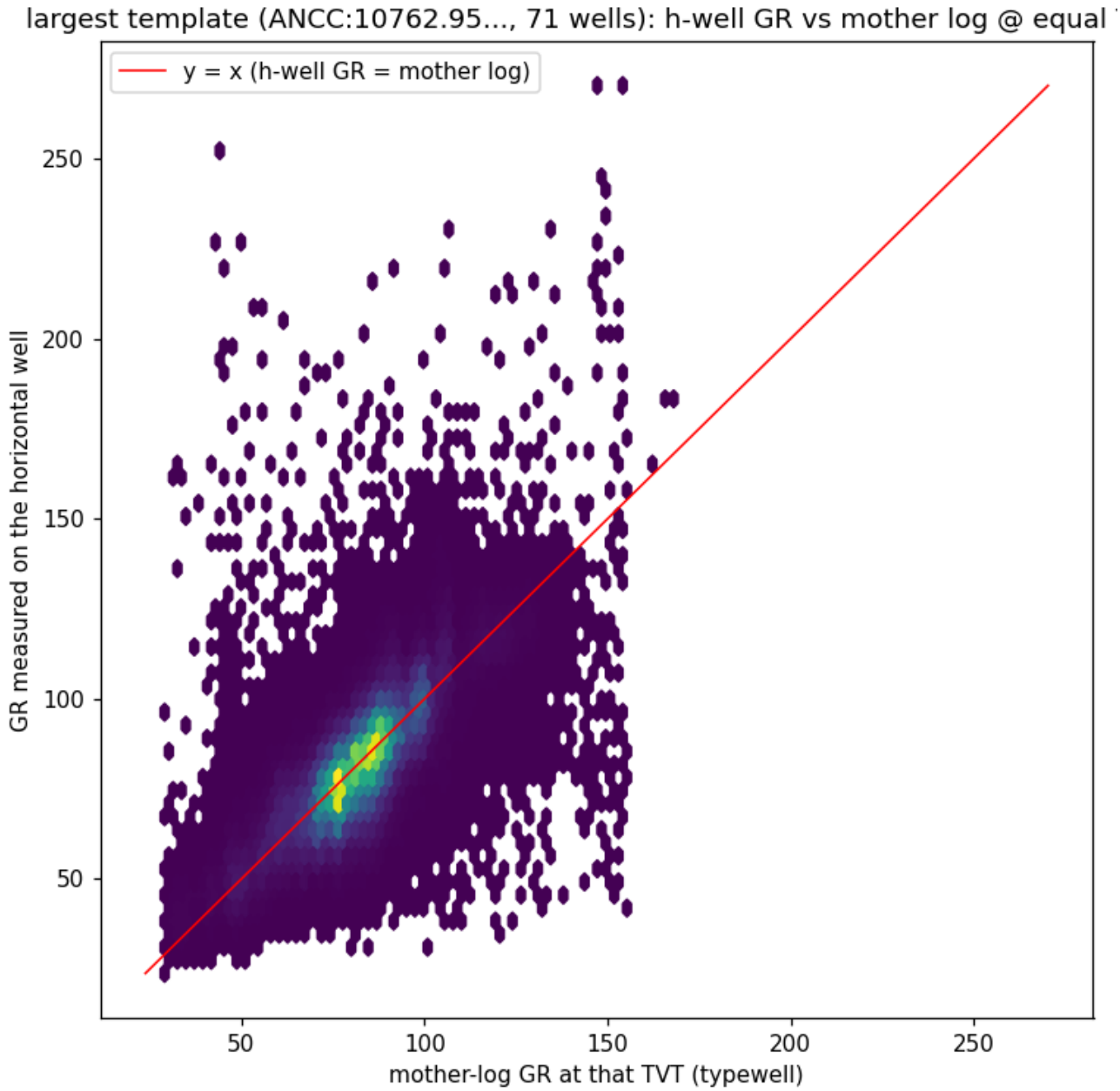
Typewell forensics: every typewell is a window of a master log

Digging into why GR matching is so ambiguous, we found the dataset's hidden structure:

- The 773 train typewells collapse to **107 formation-tops templates** and, by union-find on verbatim GR overlap, to **54 master-log systems**.
- Every typewell is a **window cut from a master GR log**: same-system typewells agree on their TVT overlap to machine precision (RMSE 0.0000 over 9,932 pairs).



- The layer-cake assumption is real but noisy: lateral GR tracks the master log at equal TVT with **corr 0.865** and **~12 API** residual scatter — enough to inform, not enough to pin TVT. That scatter is the multimodality everyone fought all competition:



- **Finding — sibling-lateral likelihood tables.** Pool the laterals of same-system wells into empirical per-2-ft-TVT-bin tables of GR **bias and σ** vs the master log (leave-one-well-out):
  - the information is real: **+0.73 nat/row** at the true depth, +1.91 margin vs decoy depths; the production PF's likelihood width was ~4× wider than the true at-depth scatter;
  - the naive exploitation — swapping the sharpened likelihood into the PF — **failed 5/5 profiles** (the broad likelihood was doing implicit robustness work);
  - what worked: (a) **bias-corrected emission + surface-OU term** → our best single PF (−0.78 at full scale), (b) **extra NN input channels** → biggest single NN screen win of the project (−0.42).

The coverage surprise → best solo 5.83

Then CV and LB decorrelated across the template family:

Version	CV	Public LB
strict match gate, v1	6.474	6.212
strict match gate, v2	6.430	6.188
strict + table bank	6.162	6.208
strict + pruning (**best CV**)	6.096	**6.363 (worst LB)**
same pool, **loose test-side gate**	6.430 (bit-identical)	**6.062**
symmetric GR-cluster retrain (v109)	—	**5.826**

- **Finding — the hidden lever was test-side match coverage, and CV is structurally blind to it.** The strict template gate matched ~90% of train wells (CV looked great) but under-matched hidden wells; loosening only the test-side gate — with bit-identical CV — moved LB by −0.13. Chasing CV past that point had been actively counterproductive.
- The proper fix was symmetric: regroup by verbatim-GR clusters (98.3% of wells matchable), retrain the NN on tables built with the same grouping → **5.826**, top-10 at the time.
- The durable lesson: **when CV and LB disagree systematically, don't average them — find the mechanism.** It became the theme of the whole competition (Part IV).

Part II — The merge: five engines → 4.90 public

Four solo lines, each already competitive on its own, merged into one team.

Blend machinery

- Every member ships cached `oof_<name>.parquet` + `test_<name>.parquet` → any candidate recipe re-scores in ~2 CPU-minutes.
- Stack = **shrink-NNLS** over 40–50 members + a heavily regularized Ridge residual layer; ranked by **cv_refit** (leave-fold-out refit of the whole stacker — in-sample stacker CV is fiction).
- Multi-seed members stay separate columns; all correlation diagnostics in **residual space** (raw-pred correlations are 0.998 by construction).

Membership rules (written in blood)

- **Rule 1 — decode-parity:** a member enters only if its test-time decode is **bit-exact** to the code path that produced its OOF (shared module, verified). Two versions died to "mispriced" members before this became a hard gate.
- **Rule 2 — solo-LB-proven only:** members justified by CV lift alone inverted on LB **four times in a row**; members with proven solo transfer blended multiplicatively:

Member	Mechanism	resid-corr to pool	ΔCV	Δpublic LB
`adil_exp151` (2-D U-Net SDF)	GR + geometry only	**0.549**	−0.414	−0.335 (0.8×)
`ochir_sub12` (physical MoE)	beam-DP + analogs	0.61	−0.191	**−0.397 (2.1×)**

Member	Mechanism	resid-corr to pool	ΔCV	Δpublic LB
4 CV-only members (no solo proof)	various	—	−0.1⋯−0.3	**+0.1…+0.5 each**

- **Finding — the Q–HMM disagreement gate.** On wells where two semi-independent channels of Alexey's engine (consistency composite "Q" vs HMM decode) disagree by ≥ 11.5 ft, the blend is probably lost → override those wells toward the members that handle reference-confusion best. CV −0.052 → **LB −0.167 (3.2× transfer)** — our strongest late lever.

Final submissions

Slot	Public	Private
CV slot — our pool line + disagreement gate	5.079	**6.543** (scored)
Mix slot — avgTW hedge line	4.940	6.664
best public sub (not selected)	4.901	6.898

Part III — The five engines

Engine	Author	One-line formulation	Solo public LB
dTVT-cumsum NN + PF/template pool	Phong	increment NN + PF/template channels; NNLS/Ridge pool	5.83
Chunked 2-D U-Net SDF	Adil	SDF over (residual-TVT × MD) chunks, MTP K=4, re-anchored rollout	6.02
Segment-residual physical MoE	Ochir	beam-DP in surface space + analogs + bounded gates	**5.28**
Typewell Editor	Andrey	NN corrects the typewell GR itself; avgTW hedge engine	5.74 (self-blend)
Static PF/Q/HMM engine	Alexey	kriging prior → PF legs → Q → HMM, zero retrain	≈ 6.0

Adil — chunked 2-D U-Net SDF with re-anchored rollout

- 9-channel 256×64 image per chunk; U-Net predicts a signed-distance field with **chunked multi-trajectory prediction (K=4) + selector**; autoregressive decode with **75%-overlap re-anchoring**.
- GR + geometry only — no formations, no PF — which is exactly why it was the most orthogonal member in the pool (resid-corr 0.549).
- Solo lineage: 6.516 → 6.238 (training through the chunk handoff) → **6.016** (anchor-point conditioning).
- **Adil's finding:** the hidden test contains **re-interpreted copies of train wells** (matchable by row count + Z + GR + prefix). Blending `0.45·train_TVT + 0.55·pred` on matched wells ≈ −0.03…−0.05 LB, essentially free.

Ochir — segment-residual physical MoE (solo LB 5.283)

- Fully physical/analog, no GBM-FE anywhere: four **beam-search DP decoders** in surface space `U = TVT + Z` (incl. a directional self-reference — a GR(TVT) template from the well's own prefix, split by drilling direction — and a cusp-shaped typewell response), a **spatial neighbor-well analog** (k-NN in heel XY + azimuth), an LGBM analog-ranker.
- **Bounded per-segment NN gates**: tanh-deltas chained to an already-validated prior simplex — the gates can nudge, never invent.
- Its local CV (7.0) understated its LB (5.28) by 1.7 ft — analog methods generalize to hidden wells better than they cross-validate. As a pool member: NNLS weight 0.33, rank #1, −0.397 public.

Andrey — the Typewell Editor (edit the reference, not the prediction)

- A residual TCN **corrects the typewell GR itself** before any decoding: robust affine fit on the prefix + bounded per-bin edits (tanh ±0.7σ, global gain/offset ±0.2).
- Loss = **measurement consistency**: the corrected typewell, sampled at the true TVT, must match the lateral GR (truth used only as a sampling target → inference is leakage-safe).
- As an accuracy lever it was flat (6.051 vs 6.052 base — GR-fit ≠ TVT-fit on average). As a **diversity lever** it was outstanding: a 50/50 self-blend of the same engine with/without editing scored **5.741 (−0.31)**, and the averaged-typewell variant became the backbone of the team's final mix line (public 4.90–4.94).

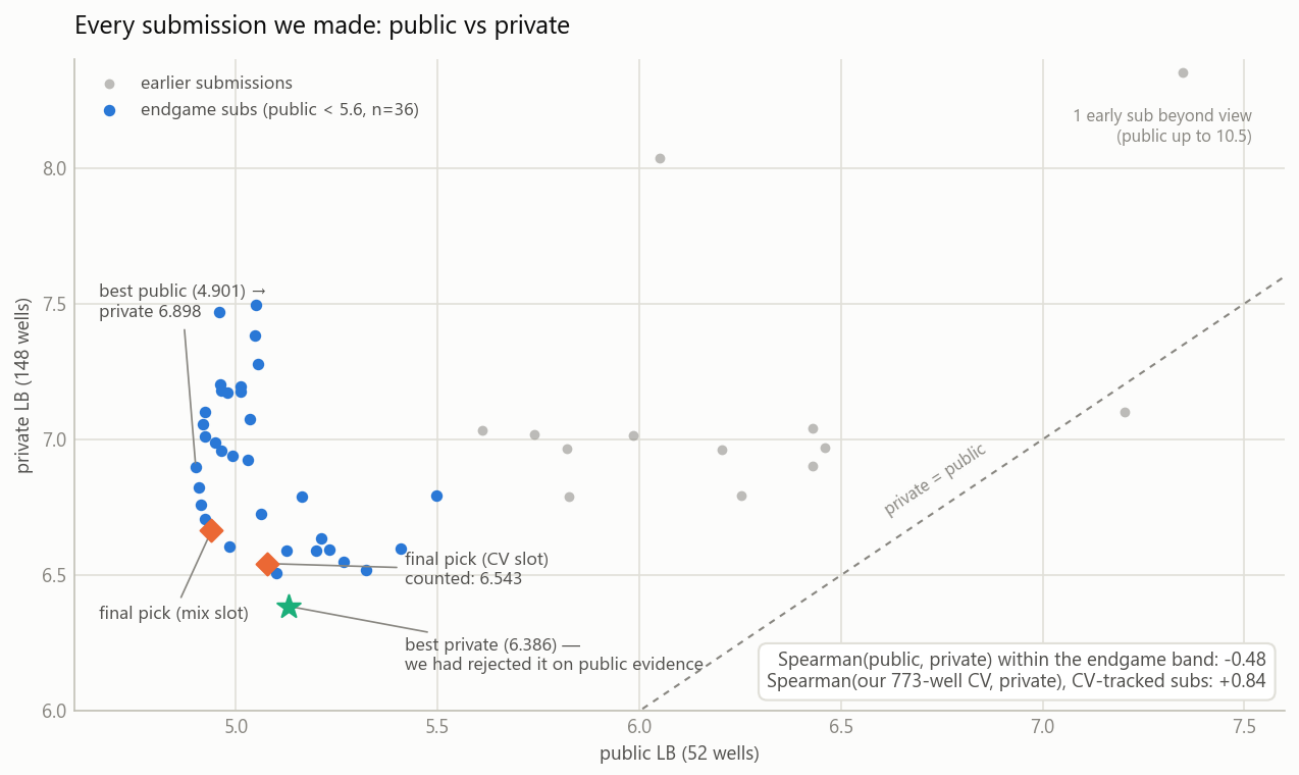
Alexey — static PF/Q/HMM engine + the GR-normalization constraint

- Zero-retrain engine: kriging surface prior from all train picks → PF (forward + quarter-backward legs) → consistency composite (Q) → HMM decode → profile combiner. Solo line ≈ LB 6.0; its Q and HMM channels power our disagreement gate and our strongest late NN members.
- **Alexey's finding** (we verified it on all 773 train wells): the host's GR normalization leaves an exact per-well invariant

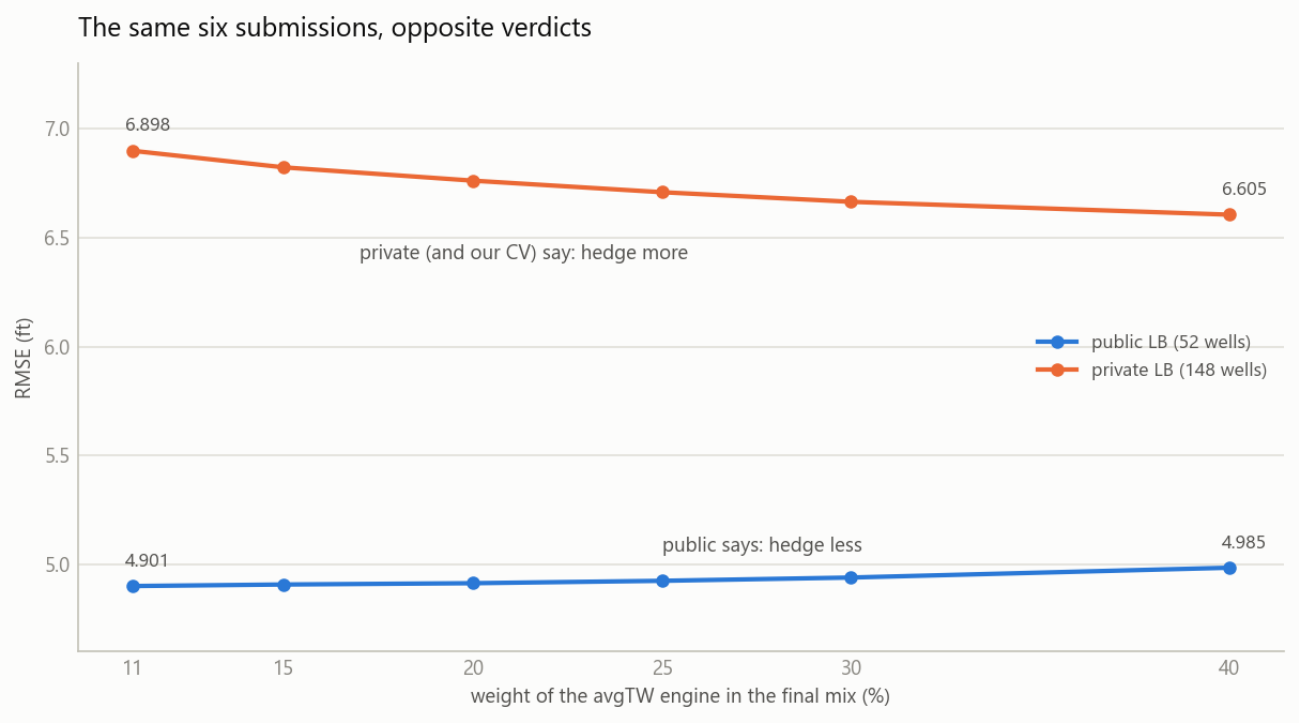
```
β = ⟨GR_lateral, TW_GR(TVT_true)⟩ / ⟨GR, GR⟩ = 0.99904 ± 0.001
```

- Given the known prefix, this pins the hidden-zone cross-term `Σ_tail GR·TW(TVT)` to **0.083 API MAE against a 17.4 API spread** — an exact, test-legal scalar constraint on the answer (99.99% of that quantity's variance).
- Neither of us fully monetized it: he used it as a GR tail rescale; our attempt to project predictions onto the constraint failed because the constraint's root set is multimodal (median 8 solutions in ±40 ft — the typewell strikes again). We still believe there was more in this one.

Part IV — What the private LB taught us



- All **50** submissions landed above the diagonal: **+1.47 ± 0.95 ft** on average (endgame band +1.8).
- Correlation ladder — the tighter the band where we were actually making decisions, the more anti-correlated public became: all 50 subs **−0.07** → endgame band (36 subs) **−0.48** → the 13 subs with tracked CV **−0.69**. Meanwhile **CV(773 wells) ↔ private = +0.84** (Pearson +0.92).
- The cleanest instance — six final-mix candidates differing only in how much edited-typewell engine they hedge with. Public ranks them one way, private exactly the other, and our CV agreed with **private**:



- Our best private score (**6.386**) was a submission we had labeled overfit-to-CV — on public-LB evidence.
- Our read (matching what the #6 and #8 writeups observed independently): the 52-well public split was not exchangeable with the private 148, and it systematically flattered exactly the method class we leaned into in the endgame — pure GR-matching signals and per-well gates tuned against public anchors. What survived on private was what our 773-well CV liked: the decode-parity members, the physical/analog line, and hedging more rather than less.
- We kept the gold because one final slot was reserved for pure CV. The rule we'll carry forward: **an honest n=773 CV outranks an n=52 public LB; when they disagree systematically, believe the CV and go hunting for the mechanism** — sometimes it's leakage, sometimes a real test-side lever (our template-gate coverage), and sometimes the public split itself.

What didn't work

Attempt	Local signal	LB	Takeaway
OOF-trained drift-direction correctors (sign / onset / midpoint)	CV −0.25	**+0.34**	correctors memorize OOF idiosyncrasies; drift direction is the info wall
Ridge residual layer extrapolating to hidden wells	CV positive	≈ **+1.0**	cut back to NNLS-only was our biggest single LB jump
Seed-enrichment of an already-strong member (×2)	CV −0.09 / −0.06	+0.14 / +0.11	NNLS trades the real member for its CV-flattered twin

Attempt	Local signal	LB	Takeaway
Learned per-well routing / MoE over members	CV −0.33	+0.33	value lives in experts, not routers (Ochir reached the same conclusion internally)
Sharpened honest-σ likelihood swapped into the PF	+0.73 nat/row	worse 5/5	a broad likelihood is implicit robustness, not a bad σ estimate
GR offset re-calibration inside the PF likelihood (×3)	—	unstable	butterfly sensitivity
U-Net on the per-row dTVT regression target	train collapse	—	the target encoding was wrong, not the architecture — see the 2-D-grid solutions above us
Feature expansion of the minimal-FE NN (14 → 52)	CV +0.49	—	its blend value was its minimalism
Synthetic wells by donor relocation (drift target copied verbatim)	pretrain - transfer +	blend ±0.00	no new (trajectory × geology) combinations by construction; the compose-style generators in #2/#8/#13 are what we should have built. Clearest miss of our campaign

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Congratulations to the winners — the top writeups are a masterclass (the probability-field formulations and synthetic-data factories in particular), and thanks again to my teammates for a merge where five very different engines actually added up.